

# GEN AI LEARNING ASSESSMENT

## Multi-Step Prompt Workflow with Chained Prompts and Critique Gate

**Student Name: Gurram Venkatesh**

**Reg No: 99230040026**

**Department: CSE AIML**

Scenario: Technical Documentation – Smart Health Device

Workflow: Raw Notes → Structured Documentation → Self-Audit & Final Refinement

### 1. Assessment Requirements

- At least 3 distinct prompts must be chained in the same chat session.
- Each later prompt must explicitly use, critique, or transform the output of the previous step.
- At least one step must contain a critique/evaluation/refinement gate with specific constraints.
- The workflow should be tested live and the outputs should be documented.
- The final submission must contain the exact prompt chain, intermediate AI responses, and direct evidence such as a chat link or screenshot.

### 2. Selected Scenario

I selected a technical-documentation workflow. The task is to transform rough product notes for a Smart Health Monitoring and Medication Reminder Device into a clear product guide, then evaluate the draft against a defined checklist and produce a corrected final version.

Why this scenario is suitable for chaining:

- The raw notes first need to be extracted and organized.
- The organized plan is then required to generate the documentation.
- The generated documentation must then pass a self-review gate before finalization.

### 3. Source Notes Used as Input

- Product name: Smart Health Monitoring and Medication Reminder Device.
- The device is intended to support people who need regular heart-rate checks and medication reminders.
- A heart-rate sensor is used to obtain a reading.
- The user places a finger correctly on the sensor.
- The device monitors the heart rate for approximately 60 seconds.
- The result is classified as low, normal, or high according to the configured thresholds.
- If the result is abnormal, the device gives an alert and advises the user to consult a doctor.
- The pill box can automatically open to support timely medication access.
- The documentation must avoid claiming that the device diagnoses a medical condition.

### 4. Step 1 – Extraction and Planning

Goal: Establish the role, analyze the raw notes, identify the important information, and create a structured documentation plan.

## **Prompt 1 – Exact Prompt Used**

Act as a senior technical writer for a consumer health-device company.

I will provide raw product notes for a Smart Health Monitoring and Medication Reminder Device.

Your task is to:

1. Extract the product purpose, main features, user workflow, alerts, and safety limitations.
2. Separate confirmed information from information that should not be invented.
3. Create a structured outline for a 700-word user-facing technical guide.
4. Use clear, simple language suitable for a general user.
5. Include sections for Overview, How It Works, Step-by-Step Use, Alerts and Results, Safety Notes, and Limitations.
6. Do not claim that the device diagnoses diseases or replaces a doctor.

RAW NOTES:

[Use the Source Notes provided above]

Output format:

- A. Key extracted facts
- B. Information that must not be invented
- C. Proposed documentation outline
- D. Writing constraints for Step 2

Do not write the full guide yet.

## **Intermediate AI Response – Step 1**

A. Key extracted facts

- The product is a Smart Health Monitoring and Medication Reminder Device.
- It uses a heart-rate sensor and requires the user to place a finger correctly.
- The device monitors heart rate for approximately 60 seconds.
- Results are classified as low, normal, or high using configured thresholds.
- Abnormal results trigger an alert and a recommendation to consult a doctor.
- The pill box can automatically open to support timely medication access.
- The documentation is intended for general users and should be easy to follow.

B. Information that must not be invented

- Exact medical thresholds are not provided and must not be created.
- Sensor accuracy, regulatory approval, clinical validation, battery life, connectivity, and medication dosage information are not provided.
- The device must not be described as diagnosing a disease or replacing professional medical advice.

C. Proposed documentation outline

1. Overview
2. How the Device Works
3. Step-by-Step Use
4. Understanding Alerts and Results
5. Medication Reminder Function
6. Safety Notes
7. Limitations

#### D. Writing constraints for Step 2

- Keep the guide approximately 700 words or less.
- Use short paragraphs and numbered steps where appropriate.
- Use simple, professional language.
- Do not invent specifications or medical thresholds.
- Include a clear safety disclaimer.

## 5. Step 2 – Generation / Drafting

Goal: Use the complete Step 1 output as the controlling plan and generate the detailed technical guide without adding unsupported specifications.

### Prompt 2 – Exact Prompt Used

Using ONLY the Step 1 analysis and outline above, write the first draft of the user-facing technical guide.

#### Requirements:

- Follow the exact section order from Step 1.
- Keep the guide at approximately 700 words or less.
- Use short paragraphs and numbered instructions.
- Explain the 60-second heart-rate reading clearly.
- Explain low, normal, and high results without inventing numerical thresholds.
- Explain what happens when an abnormal result is detected.
- Explain the automatic pill-box opening function.
- Include a safety note telling users to consult a qualified doctor for concerning or abnormal results.
- Do not say the device diagnoses, treats, or prevents a disease.
- Do not invent accuracy percentages, medical certifications, battery specifications, connectivity features, or medication dosage instructions.
- End with a short limitation statement.

Return only the first draft of the guide.

### Intermediate AI Response – Step 2

Smart Health Monitoring and Medication Reminder Device

User Guide – First Draft

#### 1. Overview

The Smart Health Monitoring and Medication Reminder Device is designed to support regular heart-rate monitoring and medication routines. It combines a heart-rate sensor with a pill-box reminder function to make daily health routines easier to follow.

#### 2. How the Device Works

To obtain a reading, the user places a finger correctly on the heart-rate sensor. The device monitors the heart rate for approximately 60 seconds and then provides a result based on its configured classification.

The result is presented as low, normal, or high. The device uses configured thresholds for these categories; specific numerical thresholds are not provided in this guide.

#### 3. Step-by-Step Use

1. Sit comfortably and prepare to take a reading.
2. Place your finger correctly on the sensor.
3. Keep your finger in position while the device monitors the heart rate.
4. Wait for approximately 60 seconds for the reading to complete.
5. Check the displayed result.
6. If the result is abnormal, follow the alert and seek appropriate professional advice.

#### 4. Understanding Alerts and Results

A normal result indicates that the reading falls within the device's configured normal category. A low or high result indicates that the reading falls outside that category.

When an abnormal result is detected, the device provides an alert and advises the user to consult a doctor. Users should not use the device result as a substitute for professional medical evaluation.

#### 5. Medication Reminder Function

The device can automatically open the pill box to support timely access to scheduled medication. The medication reminder function is intended to assist with routine adherence. Users should follow the medication instructions provided by their qualified healthcare professional.

#### 6. Safety Notes

Use the sensor as instructed and remain still during the reading. If a result is concerning, repeatedly abnormal, or accompanied by symptoms, consult a qualified doctor or seek appropriate medical attention.

Do not change medication or treatment based only on a device reading. The device should be used as a support tool and not as a replacement for professional healthcare.

#### 7. Limitations

This guide does not specify medical thresholds, sensor accuracy, clinical validation, regulatory certification, battery performance, or connectivity features because those details were not provided. The device should not be described as diagnosing, treating, or preventing a disease.

## 6. Step 3 – Critique Gate and Final Refinement

Goal: Force the AI to inspect the Step 2 draft against explicit requirements, identify failures, and then produce a corrected final version.

### Prompt 3 – Exact Prompt Used

Act as a strict senior technical editor.

Review the Step 2 draft you just produced. Do not assume that it is correct.

Evaluate it against this checklist:

1. Does it follow every section required by the Step 1 outline?
2. Is it approximately 700 words or less?
3. Does it avoid invented numerical heart-rate thresholds?
4. Does it avoid unsupported claims about accuracy, certification, connectivity, battery life, or clinical validation?
5. Does it clearly explain the 60-second reading process?
6. Does it explain low, normal, and high results without presenting them as a medical diagnosis?
7. Does it clearly explain the abnormal-result alert and doctor consultation advice?

8. Does it explain the automatic pill-box opening function?
9. Does it avoid saying that the device replaces professional healthcare?
10. Does it contain a clear limitations section?

First, provide a concise audit table with columns: Criterion, Pass/Fail, Evidence, Correction Needed.

Second, revise the Step 2 draft wherever necessary.

Third, output the FINAL POLISHED USER GUIDE only after the audit and corrections are complete.

Important: Do not add facts that were absent from the original source notes.

## Intermediate AI Response – Step 3 Audit

### Step 3 Audit

| Criterion                        | Result | Evidence                                                                    | Correction Needed |
|----------------------------------|--------|-----------------------------------------------------------------------------|-------------------|
| Required section structure       | PASS   | All major sections from Step 1 are present                                  | None              |
| Approximately 700 words or less  | PASS   | Draft is comfortably below the limit                                        | None              |
| No invented numerical thresholds | PASS   | No numerical thresholds were added                                          | None              |
| No unsupported technical claims  | PASS   | Accuracy, certification, connectivity, and battery claims were not invented | None              |
| 60-second process explained      | PASS   | The guide states that monitoring takes approximately 60 seconds             | None              |
| Low/normal/high explained safely | PASS   | Categories are described without diagnosing disease                         | None              |
| Abnormal alert and doctor advice | PASS   | Alert and consultation guidance are included                                | None              |
| Automatic pill-box function      | PASS   | Automatic opening is explained                                              | None              |
| Does not replace healthcare      | PASS   | Safety language states that it is not a replacement                         | None              |
| Limitations included             | PASS   | A dedicated limitations section is present                                  | None              |

Overall assessment: The draft satisfies the specified checklist. Minor wording refinement can make the final guide more concise and consistent.

## Final Polished AI Response – Step 3

Smart Health Monitoring and Medication Reminder Device  
Final User Guide

### 1. Overview

The Smart Health Monitoring and Medication Reminder Device supports regular heart-rate monitoring and medication routines. It combines a heart-rate sensor with an automatic pill-box function to help users follow their daily health routine.

2. How the Device Works

Place your finger correctly on the heart-rate sensor. The device monitors your heart rate for approximately 60 seconds and then provides a result using its configured categories.

The result is classified as low, normal, or high. The specific numerical thresholds used by the device are not included in this guide.

3. Step-by-Step Use

- 1. Sit comfortably before starting the reading.
- 2. Place your finger correctly on the sensor.
- 3. Keep your finger in position while the reading is taken.
- 4. Wait approximately 60 seconds for the monitoring process to finish.
- 5. Review the result provided by the device.
- 6. If the result is abnormal, follow the alert and seek appropriate professional advice.

4. Understanding Results and Alerts

A normal result means that the reading falls within the device's configured normal category. A low or high result means that the reading falls outside that category.

If an abnormal result is detected, the device provides an alert and advises the user to consult a doctor. A device reading should not be treated as a medical diagnosis.

5. Medication Reminder Function

The pill box can automatically open to support timely access to scheduled medication. This function is intended to help users follow their medication routine.

Always follow medication instructions provided by a qualified healthcare professional. Do not change medication or dosage based only on a device result.

6. Safety Notes

Use the sensor as instructed and remain still during the reading. If a result is concerning, repeatedly abnormal, or accompanied by symptoms, consult a qualified doctor or seek appropriate medical attention.

The device is a support tool and should not replace professional healthcare, medical evaluation, or treatment.

7. Limitations

This guide does not provide numerical medical thresholds, sensor accuracy, clinical-validation details, regulatory certification, battery specifications, or connectivity specifications because those details were not included in the source information. The device should not be presented as diagnosing, treating, or preventing a disease.

7. Chaining Dependency Demonstration

| Step   | Input From Previous Step             | Transformation                                                           |
|--------|--------------------------------------|--------------------------------------------------------------------------|
| Step 1 | Raw product notes                    | Extract facts, restrictions, and create the documentation outline.       |
| Step 2 | Complete Step 1 analysis and outline | Generate the first draft strictly according to the plan and constraints. |
| Step 3 | Complete Step 2 draft                | Audit the draft against a checklist and refine it into the final guide.  |

## 8. Requirement Compliance Checklist

| Requirement                       | Status / Evidence                                                     |
|-----------------------------------|-----------------------------------------------------------------------|
| 3 distinct chained prompts        | Completed – Steps 1, 2, and 3 are separate prompts.                   |
| Same chat/session dependency      | Designed for sequential execution in one AI conversation.             |
| Explicit dependency               | Step 2 uses Step 1; Step 3 audits and transforms Step 2.              |
| Critique gate                     | Step 3 contains a 10-point evaluation checklist and correction stage. |
| Exact prompts documented          | All three prompts are included verbatim in this document.             |
| Intermediate responses documented | Step 1, Step 2, and Step 3 outputs are included.                      |